

SYSTEM FOUNDRY / BUILD 001

MarginMap

True unit economics and target-price solver for small ecommerce sellers.

STATUS	PASS	PRICE	KRW 29,000
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01 Project Overview

MarginMap is a local-first browser tool that exposes the profit hidden behind marketplace fees, ad spend, shipping, expected return costs, and fixed overhead. It calculates current unit profit and algebraically solves both break-even and target-margin prices.

Project ID	2026-08-02-0006-marginmap
Build time	Single System Foundry execution
Delivery	Static web application, tests, Dockerfile, screenshot, PDF report
Data policy	No account, no persistence, no external request

02 Target Customer

Korean solo sellers and small ecommerce operators selling through marketplaces or direct-to-consumer channels, especially teams without a finance analyst or dedicated pricing system.

03 Problem Solved

Small sellers often price from product cost alone. Percentage fees grow with revenue, return costs are probabilistic, and monthly overhead is easy to omit. The resulting margin illusion can turn apparently successful sales into cash-flow loss.

04 Core Features

1. True unit profit	Combines product, inbound/outbound shipping, packaging, marketplace, payment, ads, tax reserve, returns, and fixed cost allocation.
2. Price solver	Calculates break-even and target-margin prices while accounting for fees that change with selling price.
3. CSV export	Exports the decision metrics for records, comparison, or a spreadsheet workflow.

05 How to Use

- Run `python3 -m http.server 8080 -d app` and open `localhost:8080`.
- Enter product, logistics, percentage costs, return assumptions, and monthly operation values.
- Review current profit, margin, break-even price, and recommended target price.
- Export the scenario as CSV when a reusable record is needed.

06 System Architecture

USER INPUT	VALIDATION	ECONOMICS ENGINE	RESULT + CSV
HTML numeric form	Range, type, and finite-number checks	Deterministic formulas in <code>calculator.js</code>	DOM rendering and client-side download

07 Technology

Runtime	Any modern browser; Python standard-library server for local launch
Frontend	Semantic HTML, responsive CSS, JavaScript ES modules
Tests	Node.js built-in test runner; zero package dependency
Deployment	Static hosting or included Python Alpine Docker image

08 Validated Execution Screen

Captured from a real Chromium session after HTTP 200 and DOM value assertions. Default scenario: KRW 8,215 unit profit, 21.1% margin, KRW 41,992 target price.

MarginMap

LOCAL-FIRST · NO SIGN-UP

SELLER UNIT ECONOMICS

매출이 아니라 남는 돈을 계산하세요.

플랫폼 수수료, 광고비, 배송비, 반품 기대비용, 월 고정비를 한 번에 반영해 실제 마진과 목표 판매가를 찾습니다.

01 · INPUT

초기화

상품 조건

가격과 원가

판매가

39000 원

상품 원가

14000 원

입고 배송비

800 원

출고 배송비

3200 원

포장비

700 원

반품 처리비

2500 원

비율 비용

플랫폼 수수료

10.8 %

결제 수수료

2.9 %

광고비율

7 %

세금 적립률

3 %

예상 반품률

6 %

목표 마진율

25 %

월 운영

월 주문량

180 건

월 고정비

450000 원

02 · RESULT

CSV 내보내기

마진 지도

주문 1건당 실제 이익

₩8,215

실제 마진율

21.1%

손익분기 판매가

₩28,233

목표 마진 판매가

₩41,992

예상 월 이익

₩1,478,700

주문 1건 비용

₩30,785

총비용

₩30,785

수수료-광고-세금 적립

₩9,243

반품 기대비용

₩342

고정비 배분

₩2,500

모든 금액은 주문 1건 기준이며 반품비는 기대값으로 반영됩니다.

MarginMap MVP

계산 결과는 경영 참고용이며 세무 자문이 아닙니다.

09 Test Results

CHECK	STATUS	EVIDENCE
Dependency install	PASS	npm install completed; application has zero runtime packages
Application start	PASS	Python server returned HTTP 200
Core behavior	PASS	DOM showed KRW 8,215 profit and KRW 41,992 target price
Automated tests	PASS	6 passed, 0 failed in final run
README command	PASS	Documented python3 server command used during browser check
Secret scan	PASS	No credential pattern or tracked .env file detected
PDF output	PASS	Generated, reopened, text-checked, and rendered for visual QA

Trace note: the first test run exposed two incorrect arithmetic expectations in the test fixture. The fixture was corrected and the complete suite was rerun. Production calculation logic did not require a change.

10 Security and Limitations

Security	Local-only processing; no login, cookies, storage, analytics, shell execution, or network API. Numeric fields are validated for type and bounds.
Tax limitation	The tax reserve input is a cash-flow assumption, not a tax engine. Actual VAT and income tax depend on business-specific rules and input credits.
Return model	Expected return cost currently includes outbound shipping and handling. Inventory loss and reverse shipping can be added later.
Decision scope	Results support pricing decisions but do not replace accounting or tax advice.

11 Pricing and Revenue Model

Launch offer	KRW 29,000 one-time purchase for the downloadable local-first tool
Expansion	KRW 6,900/month Seller Pro with saved products, channel presets, and scenario history
Why it can sell	A single corrected underpriced SKU can repay the tool immediately; no onboarding or integration is needed.

12 First Customer Acquisition

- Recruit 10 beta users from Korean smart-store and solo-seller communities.
- Offer a free margin audit using the tool and collect before/after pricing evidence.
- Publish a one-page 'hidden fee checklist' with MarginMap as the downloadable calculator.
- Convert the first three testimonials into a KRW 29,000 launch page and short demo video.

13 Improvement Plan

Next 1	Saved multi-product scenarios in IndexedDB without a backend
Next 2	Marketplace presets with versioned fee tables and update warnings
Next 3	Bulk CSV import for catalog-wide repricing
Next 4	Optional cloud sync, team roles, and channel-level profitability dashboard

14 Overall Evaluation

FACTOR	/10	RATIONALE
Customer problem strength	8.2	Pricing errors directly reduce cash and often remain invisible until settlement.
One-day completeness	9.0	All three features, responsive UI, tests, Docker, screenshot, and report are complete.
Sales potential	7.4	Clear ROI and low price; trust and crowded calculator alternatives are the main hurdles.
Maintenance ease	9.2	No backend, database, account, or third-party API; formulas are isolated and tested.
Differentiation	6.8	Return expectation and reverse target pricing improve on basic margin calculators.
Scalability	7.8	Can expand into catalog analysis and channel profitability without replacing the core engine.

OVERALL SCORE

8.1 / 10

Verdict: A credible, low-maintenance micro-SaaS precursor with immediate standalone value. The MVP is ready for customer interviews and paid beta validation.